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boron nitride. We can now squeeze light and, at the same time, make it propagate over significant distances through nano-scale materials.

In the future, low-loss graphene plasmons could make signal processing and computing much faster and optical sensing more efficient.

Advantages and challenges

1. PICs can allow optical systems to be made more compact and give higher performance than with discrete optical components.

2. These also offer the possibility of integration with electronic circuits to provide increased functionality.

3. One challenge to achieving this level of integration is the size discrepancy between electronic and photonic components.

4. The emerging field of nanophotonics is focused on creating ultracompact components for realising truly nano-scale photonic devices to match their electronic counterparts.

5. PICs should also be immune to the hazards of functionality losses associated with electromagnetic pulses (EMPs), though may not be immune to a high neutron flux.

6. Waveguides, device connections and passive optical components exhibit optical losses, which often need to be compensated with optical amplifiers. These are larger and more complex than electronic amplifiers based on transistors.

7. Some types of optical components can hardly be miniaturised. For these reasons, integrated optical circuits have not reached the complexity of electronic ICs.

8. However, devices of moderate complexity can still be useful, for example, for optical-fibre communications, where these can host multiple data transmitters and/or receivers, consisting of distributed feedback lasers, optical modulators, photodiodes and optical filters, in the form of arrayed waveguide gratings (AWGs).

Growing applications

The primary application for PICs

is in the area of fibre-optic communication, though applications in other fields such as biomedical and photonic computing are also possible.

AWG, which is commonly used as optical (de)multiplexers in wavelength division multiplexed (WDM) fibre-optic communication systems, is an example of a PIC that has replaced previous multiplexing schemes, which utilised multiple discrete filter elements.

Since separating optical modes is a need for quantum computing, this technology may be helpful to miniaturise quantum computers.

Another example of a PIC in wide use today in fibre-optic communication systems is the externally modulated laser (EML), which combines a distributed feedback laser diode with an electro-absorption modulator on a single InP based chip. This kind of device will be an important high-quality building block for quantum computation and communication, since atoms can store and manipulate information carried by photons.

However, atoms can also act like other kind of optical components, emitting light with almost perfect efficiency or reflecting it like a mirror. And having many atoms interacting with each other and with photons should provide some interesting experimental opportunities for physicists. The strong interplay between the optical response and large optical forces of many atomic mirrors can give rise to interesting opto-mechanical behavior, such as self-organisation.

An example of the new breed of components is a recently-proposed novel type of bandpass-plasmonic filter that uses a response similar to electromagnetically-induced transparency to achieve multichannel filtering. This allows easy control over filtering wavelengths and bandwidths for applications in wavelength-multiplexing systems for optical computing and communications in highly-integrated all-optical circuits. ●